A Look at the Ho–Lee Short Rate Model

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Abstract

This paper studies the Ho–Lee short-rate model and its application to term-structure modelling and interest-rate contingent-claim pricing. The model is calibrated to an initial term structure through a time-dependent drift, while interest-rate uncertainty is represented by a constant volatility parameter. Zero-coupon bonds and European options on zero-coupon bonds are valued analytically and numerically using Monte Carlo simulation, finite-difference methods, and a calibrated lattice. The numerical results are used to assess consistency across valuation methods and to illustrate the implications of the model's Gaussian dynamics and absence of mean reversion.

Keywords: Ho–Lee model, short-rate model, term structure, zero-coupon bond, Monte Carlo, finite differences, interest-rate derivatives.

1 Introduction

Short-rate models provide a framework for describing the evolution of interest rates and valuing securities whose cash flows depend on the term structure. One of the earliest arbitrage-free term-structure models is the Ho–Lee model introduced by Ho and Lee [1986].

The Ho–Lee model specifies the instantaneous short rate under the risk-neutral measure as a Gaussian process with a time-dependent drift and constant volatility. Unlike equilibrium short-rate models, the drift is selected so that the model is consistent with the observed initial term structure.

The model is particularly useful as a benchmark because of its analytical tractability and relatively simple calibration. Its principal limitation is the absence of mean reversion, which causes the variance of the short rate to increase without bound as the time horizon increases. The Gaussian specification also permits negative interest rates.

The purpose of this paper is to examine the theoretical and numerical properties of the Ho–Lee model. Particular attention is given to term-structure calibration, short-rate dynamics, zero-coupon bond pricing, and the valuation of a European option on a zero-coupon bond. Analytical results are compared with Monte Carlo simulation, an implicit finite-difference method, and a calibrated recombining lattice.

2 Notation

The principal notation used throughout the paper is summarised in Table 1.

Table 1: Principal notation used throughout the paper.

Symbol	Meaning
r_t	Instantaneous short rate
$\theta(t)$	Time-dependent drift
σ	Short-rate volatility
$P(t,T)$	ZCB price at time t
T	Bond maturity
T_1	Option expiry
T_2	Underlying ZCB maturity
X	Strike price
N	Notional
$\mathbb{Q}$	Risk-neutral measure

3 The Ho–Lee Model

Under the risk-neutral probability measure $\mathbb{Q}$, the instantaneous short rate follows

$$\mathrm{d}r_t = \theta(t)\,\mathrm{d}t + \sigma\,\mathrm{d}W_t^{\mathbb{Q}}. \tag{3.1}$$

Here, $\theta(t)$ is a deterministic time-dependent drift, σ is the constant short-rate volatility, and $W_t^{\mathbb{Q}}$ is a standard Brownian motion under the risk-neutral measure.

Integrating Equation (3.1) from 0 to t gives

$$r_t = r_0 + \int_0^t \theta(s)\,\mathrm{d}s + \sigma W_t^{\mathbb{Q}}.$$

Since the Brownian component is Gaussian,

$$r_t \sim \mathcal{N}\left(r_0 + \int_0^t \theta(s)\,\mathrm{d}s, \sigma^2 t\right).$$

The expectation is

$$\mathbb{E}^{\mathbb{Q}}[r_t] = r_0 + \int_0^t \theta(s)\,\mathrm{d}s,$$

while the variance is

$$\mathrm{Var}^{\mathbb{Q}}(r_t) = \sigma^2 t. \tag{3.2}$$

As shown in Equation (3.2), short-rate variance increases linearly with the time horizon. This behaviour follows directly from the absence of a mean-reversion term.

4 Term-Structure Calibration

Let $P(0,T)$ denote the time-zero price of a zero-coupon bond maturing at T. The instantaneous forward rate is

$$f(0,T) = -\frac{\partial}{\partial T}\ln P(0,T).$$

To ensure consistency with the initial term structure, the Ho–Lee drift is specified as

$$\theta(t) = \frac{\partial f(0,t)}{\partial t} + \sigma^2 t. \tag{4.1}$$

Consequently, $\theta(t)$ is determined by the shape of the initial forward curve and the assumed short-rate volatility rather than being estimated as an independent constant.

For a continuously compounded zero yield $y(0,T)$,

$$P(0,T) = \exp[-y(0,T)T].$$

In the numerical implementation, a smooth parametric zero-yield curve is used to construct the initial discount curve. The instantaneous forward curve is then obtained numerically, after which $\theta(t)$ is evaluated using Equation (4.1).

5 Zero-Coupon Bond Pricing

The time-t value of a zero-coupon bond paying one unit at maturity T is

$$P(t,T) = \mathbb{E}^{\mathbb{Q}}\left[\exp\left(-\int_t^T r_s\,\mathrm{d}s\right)\middle|\,\mathcal{F}_t\right].$$

Under the calibrated Ho–Lee model,

$$P(t,T) = \frac{P(0,T)}{P(0,t)}\exp\left[-(T-t)(r_t - f(0,t)) - \frac{1}{2}\sigma^2 t(T-t)^2\right].$$

The bond price therefore admits the exponential-affine representation

$$P(t,T) = A(t,T)e^{-B(t,T)r_t},$$
$$B(t,T) = T - t.$$

At time zero,

$$P^{HL}(0,T) = P^{market}(0,T),$$

so the initial term structure is reproduced by construction.

6 European Option on a Zero-Coupon Bond

Consider a European option expiring at T_1 on a zero-coupon bond maturing at T_2, where

$$0 < T_1 < T_2.$$

For notional N and strike price X, the call payoff at expiry is

$$C(T_1) = N\max(P(T_1,T_2) - X, 0).$$

The corresponding put payoff is

$$P_{\text{put}}(T_1) = N\max(X - P(T_1,T_2), 0).$$

At time zero, the integrated forward-bond volatility is

$$\Sigma = \sigma(T_2 - T_1)\sqrt{T_1}.$$

The time-zero call price is

$$C(0) = N\left[P(0,T_2)\Phi(d_1) - XP(0,T_1)\Phi(d_2)\right],$$

where

$$d_1 = \frac{\ln\left(\dfrac{P(0,T_2)}{XP(0,T_1)}\right) + \frac{1}{2}\Sigma^2}{\Sigma},$$
$$d_2 = d_1 - \Sigma.$$

Here, $\Phi(\cdot)$ denotes the standard normal cumulative distribution function.

The corresponding put price is

$$P_{\text{put}}(0) = N\left[XP(0,T_1)\Phi(-d_2) - P(0,T_2)\Phi(-d_1)\right].$$

Put-call parity gives

$$C(0) - P_{\text{put}}(0) = N\left[P(0,T_2) - XP(0,T_1)\right].$$

The at-the-money forward strike is

$$X_{\text{ATM}} = \frac{P(0,T_2)}{P(0,T_1)}. \tag{6.1}$$

7 Numerical Results

The numerical experiment uses a short-rate volatility of

$$\sigma = 0.01,$$

and an initial short rate of

$$r_0 = 0.03.$$

The analytical and closed-form solutions provide benchmarks for the Monte Carlo, implicit finite-difference, and calibrated lattice methods.

7.1 Ho–Lee Lattice

Figure 1 illustrates the recombining short-rate lattice generated under the Ho–Lee model. The lattice uses $\sigma = 0.01$ and is calibrated to the initial discount curve through time-dependent level shifts.

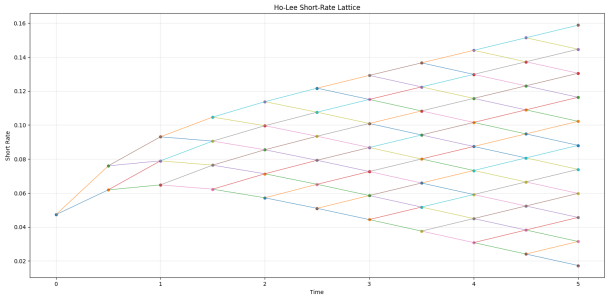


Figure 1: Calibrated recombining Ho–Lee short-rate lattice.

The lattice displays the characteristic additive structure of the Ho–Lee model. At each time step, the spacing between adjacent states is determined by the constant volatility, while the calibrated shift aligns the tree with the initial term structure.

7.2 Zero-Coupon Bond Results

A five-year zero-coupon bond with unit notional is considered. The market value implied by the initial discount curve is

$$P(0,5) = 0.67188513.$$

Table 2 compares the market value with the analytical, Monte Carlo, implicit finite-difference, and lattice estimates.

Table 2: Ho–Lee zero-coupon bond pricing results.

Method	Price	Error
Market	0.67188513	–
Analytical	0.67188513	0.000×10^{0}
Monte Carlo	0.67083218	-1.053×10^{-3}
FDM	0.66959218	-2.293×10^{-3}
Lattice	0.67188513	-1.110×10^{-16}

The analytical price reproduces the market discount factor exactly, as required by the calibration of the Ho–Lee model. The lattice value also matches the market price to machine precision because the lattice is calibrated directly to the initial discount curve.

The Monte Carlo estimate is below the market value by approximately

$$1.053 \times 10^{-3},$$

whereas the implicit finite-difference estimate differs from the market value by

$$-2.293 \times 10^{-3}.$$

The deviations in the Monte Carlo and finite-difference estimates arise from their numerical approximations rather than from the analytical term-structure calibration.

7.3 European ZCB Option Results

A European call option expiring at

$$T_1 = 2$$

years is written on a zero-coupon bond maturing at

$$T_2 = 5$$

years. The option has unit notional and uses the at-the-money forward strike defined in Equation (6.1). For the initial discount curve,

$$X = 0.77216974.$$

The closed-form Ho–Lee price provides the benchmark for the numerical valuation methods. The results are reported in Table 3.

Table 3: Ho–Lee European zero-coupon bond call option pricing results.

Method	Price	Error
Closed Form	0.01137126	–
Monte Carlo	0.01124583	-1.254×10^{-4}
FDM	0.01176308	3.918×10^{-4}
Lattice	0.01137831	7.040×10^{-6}

The lattice estimate is particularly close to the closed-form benchmark, with an absolute pricing difference of approximately

$$7.040 \times 10^{-6}.$$

The Monte Carlo estimate differs from the closed-form price by

$$-1.254 \times 10^{-4},$$

while the implicit finite-difference estimate differs by

$$3.918 \times 10^{-4}.$$

For the numerical settings used in this experiment, the lattice provides the closest approximation to the closed-form option value. The results show consistency between the analytical and numerical valuation approaches.

8 Limitations of the Ho–Lee Model

Despite its tractability, the Ho–Lee model has several important limitations.

First, the model contains no mean-reversion mechanism. As established in Equation (3.2),

$$\mathrm{Var}^{\mathbb{Q}}(r_t) = \sigma^2 t.$$

Consequently,

$$\lim_{t\to\infty} \mathrm{Var}^{\mathbb{Q}}(r_t) = \infty.$$

Short-rate uncertainty therefore increases indefinitely with the time horizon. This behaviour differs from mean-reverting Gaussian models in which the distribution of the short rate can approach a stationary long-run variance.

Second, the Gaussian specification assigns positive probability to negative short rates:

$$\Pr(r_t < 0) > 0.$$

Whether this is economically problematic depends on the application and interest-rate regime, but it remains an important structural property of the model.

Third, the model assumes a constant instantaneous volatility σ. Although this assumption contributes to analytical tractability, it restricts the model's ability to reproduce more complex volatility structures observed in interest-rate markets.

These limitations motivate extensions of the Gaussian short-rate framework that introduce mean reversion while retaining consistency with the observed initial term structure [Brigo and Mercurio, 2006].

9 Conclusion

The Ho–Lee model provides a simple and tractable arbitrage-free framework for modelling the term structure of interest rates. Its deterministic time-dependent drift allows the model to reproduce an observed initial discount curve, while its Gaussian dynamics lead to analytically tractable short-rate distributions and zero-coupon bond prices.

The paper considered the calibration of the model to an initial term structure and the valuation of a zero-coupon bond and a European option on a zero-coupon bond. Analytical results were compared with Monte Carlo simulation, an implicit finite-difference method, and a calibrated recombining lattice.

For the five-year zero-coupon bond, the analytical and lattice values reproduced the market discount factor, while the Monte Carlo and finite-difference methods produced small numerical deviations. For the European zero-coupon bond call, all three numerical methods produced values close to the closed-form benchmark, with the lattice giving the smallest pricing difference for the numerical settings considered.

The simplicity of the Ho–Lee specification is also the source of its principal limitations. In particular, the absence of mean reversion causes short-rate variance to increase linearly with time, while the Gaussian distribution permits negative rates. Nevertheless, the Ho–Lee model remains an important benchmark for understanding arbitrage-free term-structure modelling and provides a natural reference point for more sophisticated short-rate models.

References

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